Fabian N. Monrose

GEORGIA INSTITUTE OF TECHNOLOGY — *E-mail*: fabian@ece.gatech.edu
School of Electrical and Computer Engineering — *Last update*: June 17, 2026
Klaus Advanced Computing Building

Education

Ph. D., Computer Science, — *Courant Institute of Mathematical Sciences*
New York University, New York, USA — May, 1999
Advisor: Prof. Zvi Kedem

M. Sc., Computer Science, — *Courant Institute of Mathematical Sciences*
New York University, New York, USA — May, 1996

B. Sc., Computer Science, — *Barry University*
Miami, Florida, USA — May, 1993

Professional Experience

JULIAN T. HIGHTOWER CHAIR, — *Georgia Tech*
College of Engineering, — August 2022 — present

KENAN DISTINGUISHED PROFESSOR, — *University of North Carolina, Chapel Hill*
Computer Science Department, — July 2017 — July 2022

CO-FOUNDER, — *Zeropoint Dynamics*
Role: Chief Scientist, — March 2015 — 2022

DIR., COMPUTER & INFORMATION SECURITY — *Renaissance Computing Institute*
Joint appointment with UNC-CH Comp. Sci. — January 2014 — 2018

PROFESSOR, — *University of North Carolina, Chapel Hill*
Computer Science Department, — January 2013 — present

ASSOCIATE PROFESSOR, — *University of North Carolina, Chapel Hill*
Computer Science Department, — July 2008 — December, 2012

ASSOCIATE PROFESSOR, — *Johns Hopkins University*
Computer Science Department, — July 2007 —June 2008

ASSISTANT PROFESSOR, — *Johns Hopkins University*
Computer Science Department, — November 2002 — June 2007

RESEARCH SCIENTIST, — *Bell Laboratories, Lucent Technologies*
Secure Systems Research, — July 1999 — October 2002

Recent Honors

- Distinguished Reviewer, *ACM CCS Symposium*, — Fall, 2025
- CIOS Honor Roll, *Center for Teaching and Learning*, — Fall, 2023
- CSSA Graduate Teaching Award, *CS Department*, — May, 2021
- Undergraduate Students Teaching Award, *CS Department*, — May, 2015

- Best Student Paper Award, *IEEE Symposium on Security & Privacy* May, 2013
- Outstanding Research in *Privacy Enhancing Technologies* (PET), July, 2012
- AT&T Best Applied Security Paper Award, *NYU-Poly CSAW*, Nov, 2011
- Best Paper Award, *IEEE Symposium on Security & Privacy*, May, 2011
- Best Paper Award, *Intl. Conf. on Internet Monitoring and Protection*, April, 2011
- Faculty Research Award, *Google* May, 2011

Funding Awarded Grants & Contracts (*last 15 years only*)

PI, DARPA (Jim Simpson (*lead, Cynnovative*)), STTR Phase II: Learning an Unknown Compiler for Instant Development (LUCID) for $540, 000. April 2026 – May 2028.

Co-PI, ARPA-H (Brendon Saltaformaggios (*lead*)), Hospital-Integrated Vulnerability Identification and Proactive Remediation (H-VIPER) for $208, 000. October 2025 – May 2027.

Co-PI, NAVY/National Security Technology Accelerator (K. Bittick at GTRI (*lead*)), Firmware Bill of Materials Extractor for $684,000. August 2024 – July 2027.

PI, DARPA (M. Antonakakis at GaTech (*lead*)), A Novel Framework to Assess, Statistically Model and Evade Cyber Attack Infrastructure for $1, 266, 714, October. 22 — October 2025.

Co-PI (with Jan-Michel Frahm), CIA, Cyber Identity and Behavioral Analytics Research Consortium (CIBAR) for $559, 000, July 2017 — Dec, 2021.

PI, DARPA (M. Antonakakis at GaTech (*lead*)), Attributing Cyber Actors through Tensor Decomposition and Novel Data Acquisition for $2, 019, 181, Nov. 2016 — August 2021.

PI, NIST, Knowledge is Power: Assessing the Security and Privacy of Consumer IoT Architectures and Systems for $183, 063.00, July 2017 — July, 2018.

PI, IEEE, Advancing Cybersecurity Education Through Active, Challenge-Based, Learning Exercises for $119, 999.44, Jan. 2017 — August, 2018.

PI, Department of Defense University Research Instrumentation Program (DURIP), Next Generation Defenses against Web-based Exploits for $116, 500.00, April, 2016 — May 2017.

Co-PI (Jan-Michel Frahm (*lead*)), Department of Defense University Research Instrumentation Program (DURIP), Understanding Privacy Risks of Ubiquitous Personal Augmented Reality Head-mounted Displays for $95, 385.00, June, 2014 — May 2015.

PI, NSF *Secure and Trustworthy Computing*, TWC: TTP Option: Small: Scalable Techniques for Better Situational Awareness: Algorithmic Frameworks and Large Scale Empirical Analyses for $667, 900.00, Sept. 2014—August 2017.

PI Department of Homeland Security *Transition to Practice Program*, Supplement to NSF SDCI Sec: New Software Platforms for Supporting Network-wide Detection of Code Injection Attacks for $350, 000.00, Sept. 2014—August 2015.

PI, Department of Defense University Research Instrumentation Program (DURIP), Next-generation defenses for securing VoIP communications for $114, 345.00, June, 2013 — May 2014.

PI, NSF *Secure and Trustworthy Computing*, Toward Pronounceable Authentication Strings for $499, 997.00, Aug. 2013—July 2016.

PI, Department of Homeland Security, Efficient Tracking, Logging, and Blocking of Accesses to Digital Objects (C. Schmitt and M. Bailey) for $1,035,590. September 2012 — August 2014.

Co-PI (Jan-Michel Frahm (*lead*)), NSF *Division of Information & Intelligent Systems*, EAGER: Automatic Reconstruction of Typed Input from Compromising Reflections for $151,749.00, August 2011—July 2013.

PI, *Verisign Labs Research Awards Program*, On Exploring Applications of Phonetic Edit Distance for $65,000.00, June 2011.

PI, *Google Faculty Research Awards Program*, ShellOS: An Efficient Runtime Platform For Detecting Code Injection Attacks for $61,635.00, May 2011.

PI, NSF *Office of Cyberinfrastructure*, SDCI Sec: New Software Platforms for Supporting Network-wide Detection of Code Injection Attacks for $800,000.00, Aug. 2011—July 2014.

PI, NSF *Trustworthy Computing*, TC: Small: Exploring Privacy Breaches in Encrypted VoIP Communications for $496,482.00, Aug. 2010—July 2013.

PI (A. Stavrou at GMU (*lead*)), NSF *Trustworthy Computing*, Scalable Malware Analysis using Lightweight Virtualization for $259,264.00, Sept. 2009—August 2011.

Manuscripts

- *What Typos Don't Explain: An Empirical Study of Queries of Non-Existent Domain Names*. Boladji Vinny Adjibi, Michael Bailey, and Fabian Monrose. Under review at the Internet Measurement Conference (IMC), 2026.
- *Beyond Detection: Source Attribution of Synthetic Speech using Multilabel Classifier Chains*. Boo Fullwood and Fabian Monrose, under revision for re-submission to Network and Distributed Systems Security (NDSS), 2027.
- *Acoustic Microcues: Breath and Phoneme-Level Signals for Unmasking Audio Deepfakes*. Lydia Han, Amogh Palasamudram, Boo Fullwood, Fabian Monrose. In progress, 2026.
- *Exploiting Software-level Abstractions to Support Practical Hardware Trojan Attacks*. Athanasios Moschos, Georgios Kokolakis, Kevin Valakuzhy, Fabian Monrose and Angelos Keromytis, in revision for submission to Symposium on Hardware Oriented Security and Trust (HOSTS), 2027.

Publications
(*last 15 yrs only***)**

- *BRAID: Uncovering Malware Family Relationships in Data-Scarce Settings*. Kevin Valakuzhy, Miuyin Yong Wong, Omar Alrawi, Manos Antonakakis, Angelos D. Keromytis and Fabian Monrose. European Symposium on Research in Computer Security (ESORICS), September, 2026.
- *HeartBreaker: Practical Attacks for Recovering Cryptographic Keys Derived from Biosignals*. Evangelos Froudakis and Fabian Monrose. European Symposium on Research in Computer Security (ESORICS), September, 2026.
- *The Impact of Emerging AI Practices on the Cybersecurity Workforce*. Miuyin Wong, Alan F. Luo, Yunzhe Zhao, Fabian Monrose, and Michelle Mazurek. USENIX Symposium on Usable Privacy and Security (SOUPS), August, 2026.
- *Beyond the Stars: Multimodal Detection of Scams on GitHub*. Tillson Galloway, Kevin Valakuzhy, Manos Antonakakis and Fabian Monrose. USENIX Security Symposium, August, 2026.
- *SoK: Multi-Layer Indirect Call Analysis in the Real World*. Yufei Du, Vasileios P. Kemerlis, Michalis

Polychronakis and Fabian Monrose. USENIX WOOT Conference on Offensive Technologies, August, 2026.

- *The Cost of Silence: Quantifying the Prejudice of Non-Participation in Domain Name Disputes*. Boladji Vinny Adjibi, Kathryn Kleiman, Michael Bailey, and Fabian Monrose. Int. Conference on Artificial Intelligence and Law (ICAIL), June, 2026.
- *LUMEN: A Systems Approach to LLM-Guided Activation of Hidden Behaviors in Malware*. Kevin Valakuzhy, Miuyin Yong Wong, Doug Blough, Mustaque Ahamad and Fabian Monrose. International Conference on Dependable Systems and Networks (DSN), June, 2026.
- *Mind Your [M]s, Cross Your [T]s: A large-scale Phonemic Analysis of Speech Reproduction in Modern Speech Generators.* Boo Fullwood and Fabian Monrose. IEEE Conference on Acoustics, Speech and Signal Processing (ICASSP), May 2026.
- *Actively Understanding the Dynamics and Risks of the Threat Intelligence Ecosystem*. Tillson Galloway, Allen Chang, Omar Alrawi, Thanos Avgetidis, Manos Antonakakis and Fabian Monrose. Network and Distributed System Security (NDSS) Symposium, Feb 2026.
- *Repairing Trust in Domain Name Disputes Practices: Insights from a Quarter-Century's Worth of Squabbles.* Boladji Vinny Adjibi, Athanasios Avgeditis, Manos Antonakakis, Alberto Dainotti, Michael Bailey and Fabian Monrose. Network and Distributed System Security (NDSS) Symposium, Feb 2026.
- *From Concealment to Exposure: Understanding the Lifecycle and Infrastructure of APT Domains*. Athanasios Avgetidis, Boladji Vinny Adjibi, Panagiotis Kintis, Tillson Galloway, Zane Ma, Omar Alrawi, Manos Antonakakis, Roberto Perdisci, Fabian Monrose and Angelos Keromytis. International Symposium on Research in Attacks, Intrusions and Defenses (RAID), October, 2025.
- *The Guardian of Name Street: Studying Defensive Registrations for the Fortune 500*. Boladji Vinny Adjibi, Athanasios Avgetidis, Michael Bailey, Manos Antonakakis and Fabian Monrose. In Network and Distributed Systems Security (NDSS), Feb. 2025.
- *Seeing is Believing: Interpreting Behavioral Changes in Audio Deepfake Detectors Arising from Data Augmentation*. Boo Fullwood and Fabian Monrose. In 18th ACM Workshop on Artificial Intelligence and Security, October, 2025.
- *Understanding LLMs Ability to Aid Malware Analysts in Bypassing Evasion Techniques*. Miuyin Wong, Kevin Valakuzhy, Mustaque Ahamad, Douglas M. Blough, and Fabian Monrose. In Int, Conference on Multimodal Interaction (ICMI), 2024.
- *Contrasting Theory to Practice: Results from Interviews with Expert Malware Analysts*. Miuyin Wong, Matthew Landen, Frank Li, Fabian Monrose and Mustaque Ahamad. In USENIX Symposium on Usable Privacy and Security (SOUPS), August, 2024.
- *CrashTalk: Automated Generation of Precise, Human Readable, Descriptions of Software Security Bugs*. Kedrian James, Kevin Valakuzhy, Kevin Snow and Fabian Monrose. In ACM Conference and Computer and Data Security (CODASPY), June, 2024.
- *Towards Practical Fabrication Stage Attacks Using Interrupt-Resilient Hardware Trojans*. Athanasios Moschos, Fabian Monrose and Angelos Keromytis. In IEEE Int. Symposium on Hardware Oriented Security and Trust (HOST), May, 2024.
- *Improving Security Tasks Using Compiler Provenance Information Recovered At the Binary-Level*. Yufei Du, Omar Alwari, Kevin Snow, Manos Antonakakis and Fabian Monrose. In ACM Conference on Computer and Communications Security (CCS), November, 2023.
- *Stale TLS Certificates: Investigating Precarious Third-Party Access to Valid TLS Keys*. Zane Ma, Aaron Faulkenberry, Thomas Papastergiou, Zakir Durumeric, Michael D. Bailey, Angelos D.

Keromytis, Fabian Monrose and Manos Antonakakis. In Proceedings of ACM Internet Measurement Conference (IMC), Oct, 2023.

- *More Carrot or Less Stick: Organically Improving Student Time Management With Practice Tasks and Gamified Assignments*. Mac Malone and Fabian Monrose. In Proceedings of ACM Conference on Innovation and Technology In Computer Science Education, July, 2023.
- *Securely Autograding Cybersecurity Exercises Using Web Accessible Jupyter Notebooks*. Mac Malone, Yicheng Wang and Fabian Monrose. In Proceedings of ACM SIGCSE Technical Symposium, March, 2023.
- *Beyond the Gates: An Empirical Analysis of HTTP-managed password stealers and operators*. Omar Alrawi, Athanasios Avgetidis, Kevin Valakuzhy, Charles Lever, Paul Burbage, Angelos Keromytis, Fabian Monrose, and Manos Antonakakis. In Proceedings of the USENIX Security Symposium, August 2023.
- *View from Above: Exploring the Malware Ecosystem from the Upper DNS Hierarchy*. Aaron Faulkenberry, Athanasios Avgetidis, Zane Ma, Omar Alrawi, Charles Lever, Panagiotis Kintis, Fabian Monrose, Angelos Keromytis, and Manos Antonakakis. In Proceedings of the Annual Computer Security Applications Conference (ACSAC), Dec 2022.
- *Automatic Recovery of Fine-grained Compiler Provenance Information at the Binary Level*. Yufei Du, Ryan Court, Kevin Z. Snow and Fabian Monrose. USENIX Annual Technical Conference, July, 2022.
- *Disentangling Style and Content for Low Resource Video Domain Adaptation: A Case Study on Keystroke Inference Attacks*. John Lim, Jan-Michael Frahm and Fabian Monrose. In ACM Conference on Data and Application Security (CODASPY), April, 2022.
- *An Online Gamified Learning Platform for Teaching Cybersecurity and More*. Mac Malone, Yicheng Wang and Fabian Monrose. ACM Annual Conference on IT Education (SIGITE), October, 2021.
- *DynPTA: Combining Static and Dynamic Analysis for Practical Selective Data Protection*. Tapti Palit, Jarin Firose Moon, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2021.
- *Applicable Micropatches and Where to Find Them: Finding and Applying New Security Hot Fixes to Old Software*. Mac Malone, Yicheng Wang, Kevin Snow and Fabian Monrose. IEEE International Conference on Software Testing, Verification and Validation, April, 2021.
- *The Circle Of Life: A Large-Scale Study of The IoT Malware Lifecycle*. Omar Alrawi, Charles Lever, Kevin Valakuzhy, Ryan Court, Kevin Snow, Fabian Monrose and Manos Antonakakis. USENIX Security Symposium, August, 2021.
- *To Gamify or Not? On Leaderboard Effects, Student Engagement, and Learning outcomes in a Cybersecurity Intervention*. Mac Malone, Yicheng Wang, Kedrian James, Murray Anderegg, Jan Werner and Fabian Monrose. ACM SIGCSE Technical Symposium, 2021.
- *A Flexible Framework for Expediting Bug Findings by Leveraging Past (Mis-)Behavior to Discover New Bugs*. Sanjeev Das, Kedrian James, Jan Werner, Manos Antonakakis, Michalis Polychronakis and Fabian Monrose. ACM Annual Computer Security Applications Conference (ACSAC), December 2020.
- *Methodologies for Quantifying (Re-)randomization Security and Timing under JIT-ROP*. Md Salman Ahmed, Ya Xiao, Kevin Z. Snow, Gang Tan, Fabian Monrose and Danfeng Yao. In Proceedings of the ACM Conference on Computer and Communications Security, October, 2020.
- *Mitigating Data Leakage by Protecting Memory-resident Sensitive Data*. Tapti Palit, Fabian Monrose and Michalis Polychronakis. In Proceedings of the Annual Computer Security Application

Conference, December 2019.

- *The SEVerESt Of Them All: Inference Attacks Against Secure Virtual Enclaves*. Jan Werner, Joshua Mason, Manos Antonakakis, Michalis Polychronakis and Fabian Monrose. In Proceedings of the ACM Asia Conference on Computer and Communication Security, July 2019.
- *SoK: The Challenges, Pitfalls, and Perils of Using Hardware Performance Counters for Security*. S. Das, J. Werner, M. Antonakakis, M. Polychronakis and F. Monrose. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2019.
- *SoK: Security Evaluation of Home-Based IoT Deployments*. O. Alrawi, C. Lever, M. Antonakakis and F. Monrose. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2019.
- *Security Concerns in Asynchronous Web Server Architectures: When Performance Optimizations Empower Data-oriented Attacks*. M. Morton, J. Werner, K. Z. Snow, M. Antonakakis, M. Polychronakis and F. Monrose. In Proceedings of the IEEE European Symposium on Security and Privacy, April, 2018.
- *Practical Attacks Against Graph-based Clustering*. Y. Chen, Y. Nadji, A. Kountouras, F. Monrose, R. Perdisci, M. Antonakakis, N. Vasiloglou. In Proceedings of the ACM Conference on Computer and Communications Security, October, 2017.
- *Keeping Zombie Gadgets at Bay by Re-randomizing after Disclosure*. Micah Morton, Forrest Li, Kevin Snow, Michalis Polychronakis and Fabian Monrose. In Proceedings of Engineering Secure Systems and Software conference (ESSoS), July, 2017.
- *Revisiting Browser Security in the Modern Era: New Data-only Attacks and Defenses*. Roman Rogowski, Micah Morton, Forrest Li, Kevin Z. Snow, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE European Symposium on Security and Privacy, March 2017.
- *Virtual U: Defeating Face Liveness Detection by Building Virtual Models From Your Public Photos*. Yi Xu, True Price, Jan-Michael Frahm and Fabian Monrose. In Proceedings of the USENIX Security Symposium, August 2016.
- *Return to the Zombie Gadgets: Undermining Destructive Code Reads via Code-Inference Attacks*. Kevin Z. Snow, Roman Rogowski, Jan Werner, Hyungjoon Koo, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2016.
- *No-Execute-After-Read: Preventing Code Disclosures in Commodity Software*. Jan Werner, George Baltas, Rob Dallara, Nathan Otterness, Kevin Snow, Fabian Monrose and Michalis Polychronakis. In Proceedings of the ACM Asia Conference on Computer and Communication Security, May 2016.
- *Detecting Malicious Exploit Kits using Tree-based Similarity Searches*. Teryl Taylor, Xin Hu, Ting Wang, Jiyong Jang, Marc Stoeckin, Fabian Monrose and Reiner Sailer. In Proceedings of the ACM Conference on Data and Application Security and Privacy, pages 255-266, March, 2016.
- *Cache, Trigger, Impersonate: Enabling Context-Sensitive Honeyclient Analysis On-the-Wire*. Teryl Taylor, Kevin Snow, Nathan Otterness and Fabian Monrose. In Proceedings of the 23^{nd} ISOC Network and Distributed Systems Security Symposium (NDSS), Feb., 2016.
- *Isomeron: Code Randomization Resilient to (Just-in-Time) Return-Oriented Programming*. Luca Davi, Christopher Liebchen, Ahmad-Reza Sadeghi, Kevin Z. Snow, and Fabian Monrose. In Proceedings of the 22^{nd} ISOC Network and Distributed Systems Security Symposium (NDSS), Feb., 2015.
- *Watching the Watchers: Inferring TV Content from Outdoor Light Effusions*. Yi Xu, Jan-Michael Frahm and Fabian Monrose. In Proceedings of the 21st ACM Conference on Computer and Communications Security (CCS), November, 2014.

- *Emergent Faithfulness to Morphological and Semantic Heads in Lexical Blends*. Katherine Shaw, Elliott Moreton, Andrew White and Fabian Monrose. In Proceedings of the Annual Meeting on Phonology, February, 2014.
- *Security and Usability Challenges of Moving-Object CAPTCHAs: Decoding Codewords in Motion*. Yi Xu, Gerardo Reynaga, Sonia Chiasson, Jan-Michael Frahm and Fabian Monrose. IEEE Transactions on Dependable and Secure Computing, February, 2014.
- *On the Privacy Risks of Virtual Keyboards: Automatic Reconstruction of Typed Input from Compromising Reflections*. Rahul Raguram, Andrew White, Yi Xu, Jan-Michel Frahm, Pierre Georgel, and Fabian Monrose. IEEE Transactions on Dependable and Secure Computing, Volume 10, Issue 3, pages 154-167, May-June, 2013.
- *Seeing Double: Reconstructing Obscured Typed Input from Repeated Compromising Reflections*. Yi Xu, Jared Heinly, Andrew M. White, Fabian Monrose and Jan-Michael Frahm. In Proceedings of the 20th ACM Conference on Computer and Communications Security (CCS), November, 2013.
- *Check my profile: Leverage static analysis for fast and accurate detection of ROP gadgets*. Blaine Stancill, Kevin Snow, Nathan Otterness, Fabian Monrose, Lucas Davi, and Ahmad- Reza Sadeghi. In Proceedings of the 16th International Symposium on Research in Attacks, Intrusions, and Defenses, October, 2013.
- *Crossing the Threshold: Detecting Network Malfeasance via Sequential Hypothesis Testing*. Srinivas Krishnan, Teryl Taylor, Fabian Monrose and John McHugh. In Proceedings of the 42^{nd} Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Performance and Dependability Symposium, June, 2013.
- *Just-In-Time Code Reuse: On the Effectiveness of Fine-Grained Address Space Layout Randomization*. Kevin Snow, Lucas Davi, Alexandra Dmitrienko, Christopher Liebchen, Fabian Monrose and Ahmad-Reza Sadeghi. In Proceedings of 34^{th} IEEE Symposium on Security and Privacy, May, 2013. (**Best Student Paper Award**).
- *Clear and Present Data: Opaque Traffic and its Security Implications for the Future*. Andrew White, Srinivas Krishnan, Michael Bailey, Fabian Monrose and Phil Porras. In Proceedings of the 20^{th} ISOC Network and Distributed Systems Security Symposium, Feb., 2013.
- *Security and Usability Challenges of Moving-Object CAPTCHAs: Decoding Codewords in Motion*. Yi Xu, Gerardo Reynaga, Sonia Chiasson, Jan-Michael Frahm, Fabian Monrose and Paul van Oorschot. In Proceedings of the 21^{th} USENIX Security Symposium, August, 2012.
- *Toward Efficient Querying of Compressed Network Payloads*. Teryl Taylor, Scott E. Coull, Fabian Monrose and John McHugh. In USENIX Annual Technical Conference, June, 2012.
- *Trail of Bytes: New Techniques for Supporting Data Provenance and Limiting Privacy Breaches*. Srinivas Krishnan, Kevin Snow, and Fabian Monrose. IEEE Transactions on Information Forensics and Security, pages 1876-1889, Issue 6, Volume 7, 2012.
- *Understanding Domain Registration Abuses (Invited Paper)*. Scott Coull, Andrew White, Ting-Fang Yen, Fabian Monrose and Michael K. Reiter. Special Issue of Computers & Security (COSE), pages 806-815, Volume 31, Number 7, October, 2012.
- *iSpy: Automatic Reconstruction of Typed Input from Compromising Reflections*. Rahul Raguram, Andrew White, Dibyendusekhar Goswami, Fabian Monrose and Jan-Michael Frahm. In Proceedings of the 18^{th} ACM Conference on Computer and Communications Security (CCS), November, 2011.
- *ShellOS: Enabling Fast Detection and Forensic Analysis of Code Injection Attacks*. Kevin Snow, Srinivas Krishnan, Fabian Monrose and Niels Provos. In Proceedings of the 20^{th} USENIX

Security Symposium, August, 2011.

- *An Empirical Study of the Performance, Security and Privacy Implications of Domain Name Prefetching*. Srinivas Krishnan and Fabian Monrose. In Proceedings of the 41^{st} Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Dependable Computing and Communications Symposium, June, 2011.
- *Amplifying Limited Expert Input to Sanitize Large Network Traces*. Xin Huang, Fabian Monrose, and Michael K. Reiter. In Proceedings of the 41^{st} Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Performance and Dependability Symposium, June, 2011.
- *Phonotactic Reconstruction of Encrypted VoIP Conversations*. Andrew White, Kevin Snow, Austin Matthews and Fabian Monrose. In Proceedings of 32^{nd} IEEE Symposium on Security and Privacy, May, 2011. (**Best Paper Award**).
- *Towards Optimized Probe Scheduling for Active Measurement Studies*. Daniel Kumar, Fabian Monrose and Michael K. Reiter. In Proceedings of the 6^{th} International Conference on Internet Monitoring and Protection (ICIMP), March, 2011 (**Best Paper Award**).
- *On Measuring the Similarity of Network Hosts: Pitfalls, New Metrics, and Empirical Analyses*. Scott Coull, Micheal Bailey and Fabian Monrose. In Proceedings of the 18^{th} Annual Network and Distributed Systems Security Symposium, February, 2011.

Recent Service
(*last 5 years***)**

- ECE PhD Admissions committee (**Chair** for Software & Systems TIG), 2023-2026
- 32^{nd} ACM Conference on Computer and Communications Security, 2026
- 47^{th}IEEE Symposium on Security and Privacy, 2026
- 32^{nd} ACM Conference on Computer and Communications Security, 2025
- 14^{th} ACM Conference on Data and Application Security and Privacy, 2024
- Faculty Recruiting Committee member, School of Cybersecurity & Privacy, 2023
- 13^{th} ACM Conference on Data and Application Security and Privacy, 2023

Recent Patents

- Kevin Valakuzhy, Miuyin Yong Wong,Fabian Monrose, Douglas Blough and Mustaque Ahamda. LLM-Enabled System for Locating and Bypassing Evasive Malware Tactics, Provision Patent Filing 165972-00196, October 14, 2025.
- Kevin Valakuzhy, Miuyin Yong Wong, Omar Alrawi, Manos Antonakakis, Angelos D. Keromytis and Fabian Monrose. Systems and Methods for Malware Similarity via Datastore of Assembly-level Malicious Behaviors Extracted from Memory, US Patent Application No. 19/380,650, November, 2025.
- Jan Werner, Rob Dallara, George Baltas, Michalis Polychronakis, Kevin Snow and Fabian Monrose. *Methods, Systems, and Computer Readable Media for Preventing Code Reuse Attacks.* U.S. Patent No. 10,628,589, April, 2020.
- Xin Hu, Jiyong Jang, Fabian Monrose, Marc Philippe Stoecklin, Teryl Taylor and Ting Wang. *Detecting Web Exploit Kits by Tree-based Structural Similarity Search.* U.S. Patent 10,560,471, Feb, 2020.
- T. Taylor, K. Snow, N. Otterness, and F. Monrose. *Methods, systems, and computer readable media*

for detecting malicious network traffic, U.S. Patent No. 9,992,217, June 2018.

- Srinivas Krishnan, Fabian Monrose, Michael Bailey, Phillip Porras, and Andrew White. *Methods, Systems, and Computer Readable Media for Rapid Filtering of Opaque Data Traffic*, U.S. Patent No. 9,973,473, May 2018.
- S. Krishnan, T. Taylor, F. Monrose and J. McHugh. *Methods, Systems, and Computer Readable Media for Detecting Compromised Computing Hosts*. U.S. Patent No. 9,934,379, April 2018.
- S. Krishnan, K. Snow, and F. Monrose. *Methods, Systems, and Computer Readable Media for Efficient Computer Forensic Analysis and Data Access Control*. U.S. Patent No. 9,721,089, August 2017.
- Keven Z. Snow, Fabian Monrose and Srinivas Krishnan. *Methods, Systems, and Computer Readable Media for Detecting Injected Machine Code*. U.S. Patent 9,305,165, April 5, 2016.
- Fabian Monrose, Teryl Taylor, Srinivas Krishnan and John McHugh. United States Patent Application Serial No. 14/773,660 for *Methods, Systems, and Computer Readable Media for Detecting a Compromised Computing Host*, March, 2015.